Patterns for Prediction:

An Audio-to-Audio HMM Framework

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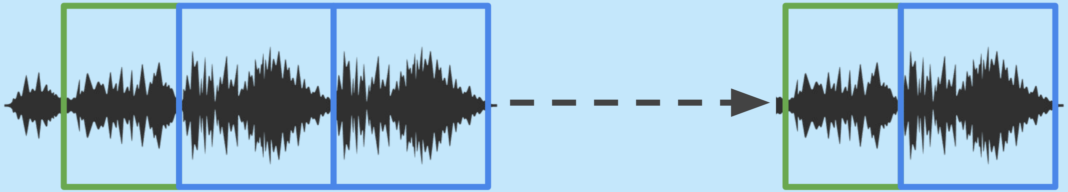
Task Description

- **What the task is:** predicting a continuation from an audio prime
- **The evaluation uses:** Cardinality Score and Pitch Score
- **The Patterns for Prediction Development Dataset (PPDD)**:
  monophonic audio → HMM model → audio → MIDI (evaluation)

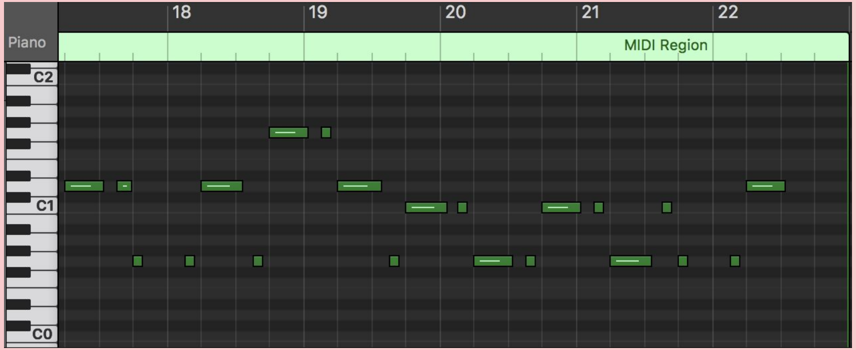


Input Prime

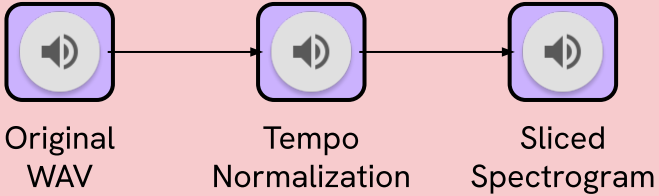
Predicted Continuation

PPDD Dataset and Preprocessing

- **Small monophonic subset of the PPDD dataset**
- **Primes have varying tempo:** Normalized all to 120 BPM
- **Resynthesis using a fixed soundfont:** uniform timbre for feature extraction
- **Beat tracking validation**: Ensures consistent tempo
- **Spiced all audio to 16th note duration:** keep each observation the same dimension (18220 observations)
- **Final dataset:** 75 verified primes → 60 training, 15 testing

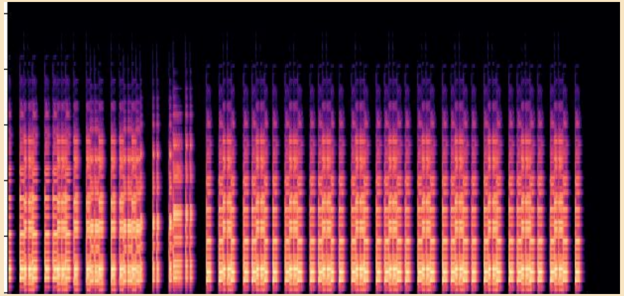


Example of one prime in PPDD



Attempt 1: Raw Spectro + HMM

- All Noise
- Unable to provide meaningful prediction



Sliced Spectrogram

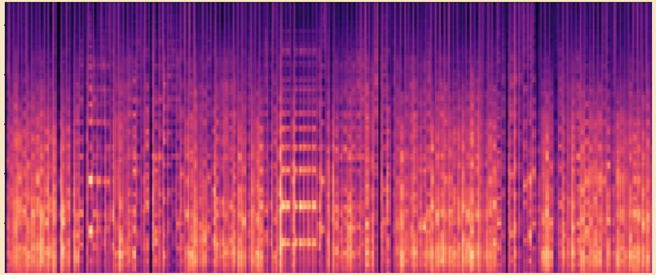
Optional PCA
Dimension Reduction

train

First-order Gaussian HMM

predict

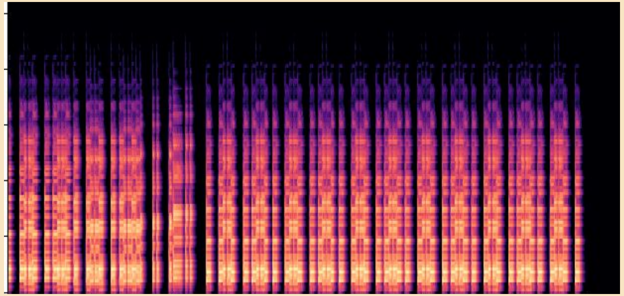
Sequence of Spectrograms slices



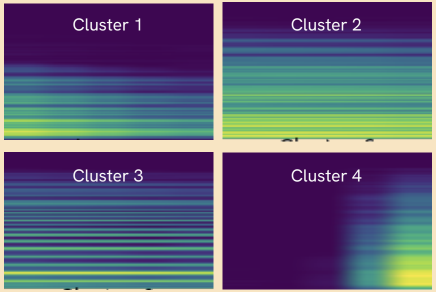
All Noise

Attempt 2: Clustering + HMM

- Start to hear melody
- Prone to noise
- Low fidelity



Sliced Spectrogram



K-Means Clustering

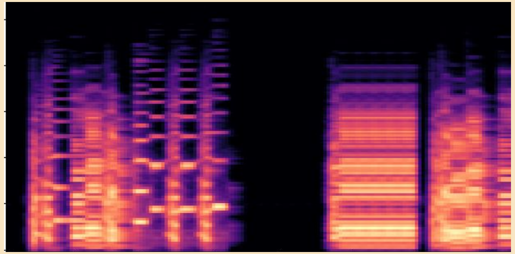
train

First-order Categorical HMM

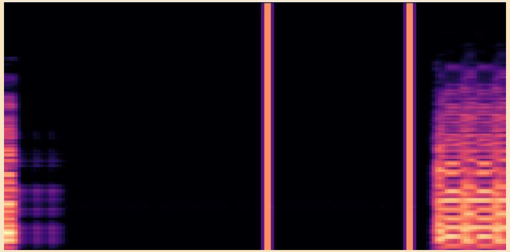
predict

Sequence of **Cluster IDs**

convert (cluster Mean)



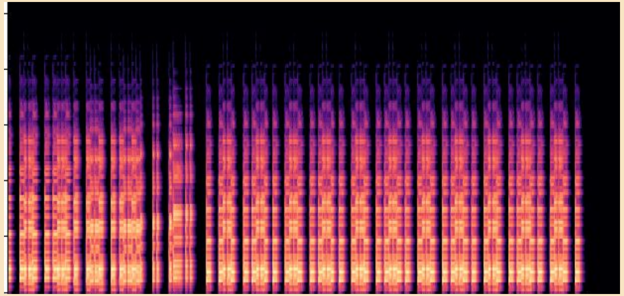
Sometimes work a little



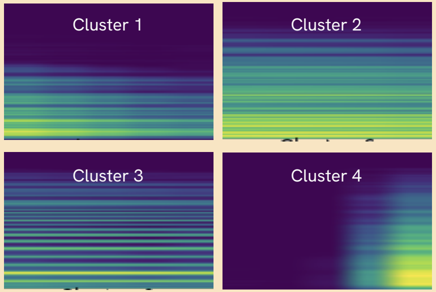
Sometimes not

Attempt 3: Variable-Order HMM

- No more noise
- Low fidelity
- Long-term structure



Sliced Spectrogram



K-Means Clustering

train

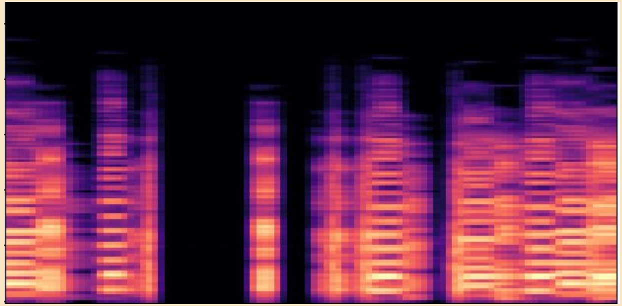
Variable-order HMM

- **Longer context**
- **Local weighting**

predict

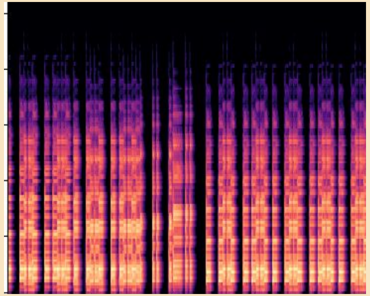
Sequence of Cluster IDs

convert (cluster Mean)



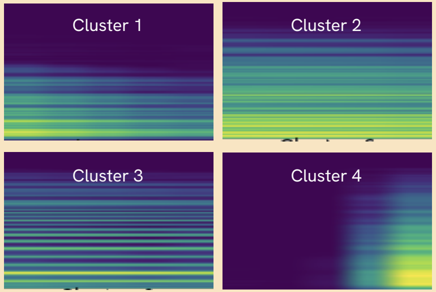
- Better fidelity
- Patterns emerge

Attempt 4: Musical Features



Sliced Spectrogram

Feature Engineering
- **Chroma**
- **RMS Energy**
- **Pitch Center**



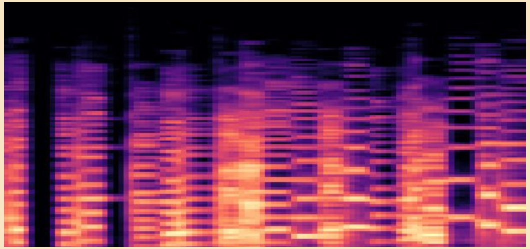
Feature Clustering Spectrogram

Variable-order HMM

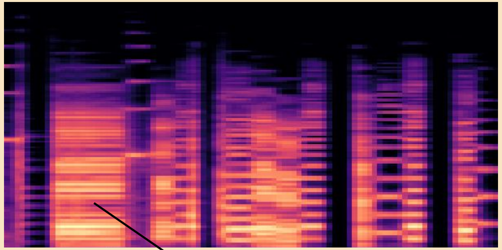
predict

Sequence of Cluster IDs

convert (cluster **example**)



musical patterns emerge



Long note

Evaluation & Sample Outputs

- Task 1: match the algorithmic output with the original continuation and compute a **match score**
  - The Cardinality Score
    - How many of the expected notes did the model generate correctly?
  - The Pitch Score
    - How close are the generated pitches to the correct pitches?
- Results on Attempt 4 (Best Performing Model)

	Attempt 4	Baseline Mirex Markov
Cardinality Score	0.263 ± 0.208	~0.27-0.30
Pitch Score	0.429 ± 0.196	~0.45-0.50

Strengths

- Fully audio-based system 🔊
- Simple, relatively interpretable, reproducible 💡
- Beat-synchronous features improve temporal consistency ⏳

Limitations

- Beat Slicing artifacts 🔬
- Spectrogram slicing → generation favored short notes 🎵
- Small, monophonic, piano dataset 🎹
- Need for tempo normalization 📏
- Evaluation metrics 📊

Conclusion

- Explored a complete **audio-to-audio** continuation system for the Patterns for Prediction MIREX task
- Musical Features **K-means Clustering** + **Variable order HMM** provide a tractable pipeline
- Establishes a baseline and opens the door for more advanced audio continuation methods

Future Work

- Integrate **neural networks** or **diffusion models** for higher fidelity waveform reconstruction
- Expand to **polyphonic** or **multi-instrument continuation** and more diverse datasets
- Incorporate **evaluation beyond MIREX**, including perceptual and musicality-centered measures
- Move toward **interactive** or **real-time continuation systems** for human-machine musical dialogue


Thank You